

Polina Dolgova

PhD Student, CISPA Helmholtz Center for Information Security

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Research Interests

- Machine unlearning, including certified guarantees
- Stability and efficiency of learning algorithms
- Mathematical foundations of machine learning
- Theory–practice connections

Publications and Preprints

Sequential Subspace Noise Injection Prevents Accuracy Collapse in Certified Unlearning.

Polina Dolgova, Sebastian U. Stich.

arXiv preprint, 2026. [arXiv:2601.05134](#)

Education

Saarland University

Jul 2025 – present

(CISPA Helmholtz Center for Information Security)

PhD in Computer Science

Saarbrücken, Germany

Supervisor: [Sebastian U. Stich](#)

The National Research University Higher School of Economics (HSE)

2021 – 2023

Moscow, Russia

M.S. in Applied Mathematics and Computer Science

Thesis: Topological Data Analysis in Medical Applications

Supervisor: [Vsevolod Chernyshev](#)

Yandex School of Data Analysis

2019 – 2022

Moscow, Russia

M.S.-level program in Machine Learning

The National Research University Higher School of Economics (HSE)

2016 – 2020

Moscow, Russia

B.S. in Mathematics (with honors)

Thesis: Clique-Chromatic Numbers for Intersection Graphs

Supervisor: [Andrei M. Raigorodskii](#)

Selected Industry Experience

Yandex — ML Engineer

2021 – 2024

- Large-scale text generation with LLMs (RL-based training, PEFT)
- Efficiency-oriented modeling under strict latency and memory constraints
- Improving diversity and controllability of generated text (objectives and generation strategies)
- Mentoring and research supervision of junior engineers and interns

Insud Pharma — ML Engineer

2024 – 2025

- Applied ML research in audio and computer vision for pharma workflows

- Data-efficient ASR adaptation under domain shift (specialized terminology, accented speech)
- OCR and anomaly detection in low-data / weakly supervised settings for manufacturing.

Talks and Posters

- Poster (workshop): *Sequential Subspace Noise Injection Prevents Accuracy Collapse in Certified Unlearning*, PPML @ EurIPS 2025. [arXiv:2601.05134](#)

Research Visits

University of Zurich (UZH), Zurich, Switzerland
Hosted by [Anastasia Koloskova](#)

Nov 2025

Teaching and Service

Coach in Olympiad Mathematics

2017 – 2024

Languages

Russian (native), English (fluent), Spanish (basic conversational)